Probstat 11

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# Distribusi Normal

- Salah satu distribusi peluang v/ variabel random kontinyu dgn domain $(-\infty, \infty)$

- Contoh:

  $X$ = nilai UTS mhs

  $0 \leq x \leq 100$

  $Y$ : berat bersih minuman kaleng

  $0 \leq y \leq 600$

Base form dist. normal for $X$ with mean $\mu$ & varians $\sigma^2$ =

$$X \sim N(\underbrace{\mu, \sigma^2}_{\text{Parameter}})$$

- Fungsi peluang dist. norm

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$\hookrightarrow P(X \leq x) = \int_{-\infty}^{x} f(x)\,dx \Rightarrow \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$